# Year 7 Mathematics

## Unit 4: Fractions

4.3 Fractions, Decimals and Percentages

# Fractions, Decimals and Percentages

STARTER · 5 MIN

**Date:** _____ **Title: Fractions, Decimals and Percentages**

**LEARNING OBJECTIVES**

- To be able to convert between fractions, decimals and percentages

**STARTER** *Answer in your book. No calculators.*

1. $\frac{1}{5} = \frac{\square}{10}$

2. $\frac{3}{5} = \frac{\square}{10}$

3. $7 \div 8$

4. $9 \div 20$

# Starter — Answers

STARTER

*Tick or correct in green pen*

1. **2**

2. **6**

3. **0.875**

4. **0.45**

# Success Criteria

SUCCESS CRITERIA

*By the end of this lesson:*

| | | |
|---|---|---|
| BRONZE | I can recognise equivalent fractions, decimals and percentages | ☐ |
| SILVER | I can divide the numerator by the denominator to find a decimal | ☐ |
| GOLD | I can convert mixed numbers to decimals and percentages | ☐ |

*Tick each box in the plenary when you can do it on your own.*

# Introduction — Fractions, Decimals and Percentages

INTRODUCTION · 3 MIN

Fractions, decimals and percentages are three ways to write the same value.

## Percent

Means 'out of 100'.

70% = 70 out of 100

## Fraction → decimal

Make the bottom 10 or 100.

$$\frac{3}{4} = \frac{75}{100} = 0.75$$

## Decimal → %

Multiply by 100.

0.75 = 75%

## Over 100%

1 whole = 100%.

$$1\ \frac{1}{4} = 1.25 = 125\%$$

# Examples 1 and 2 — Tenths and Hundredths

EXAMPLES · 5 MIN

BRONZE

*Watch and copy into your book*

**Hint** Percent: out of 100 Make the bottom 100

## Example 1

Write as a decimal and a percentage

$$\frac{7}{10}$$

## Example 2

Write as a decimal and a percentage

$$\frac{2}{10}$$

# Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

*Check your copy and correct in green pen*

## Example 1

**Make the bottom 100**

$$\frac{7}{10} = \frac{70}{100}$$

Decimal = 0.70 = 0.7

**Percentage = 70%**

Tip: 70 out of 100 = 70%.

## Example 2

**Make the bottom 100**

$$\frac{2}{10} = \frac{20}{100}$$

Decimal = 0.2

**Percentage = 20%**

Tip: 20 hundredths = 0.20.

# Task 1

PRACTICE · 9 MIN

BRONZE

*Write as a decimal and a percentage*

**Hint** Percent: out of 100 Make the bottom 100

1 $\frac{1}{10}$

2 $\frac{3}{10}$

3 $\frac{9}{10}$

4 $\frac{1}{2}$

5 $\frac{1}{4}$

★ $\frac{3}{4}$

# Task 1 — Answers

CORRECTIONS

BRONZE

*Tick or correct in green pen*

1. 0.1 = 10%
2. 0.3 = 30%
3. 0.9 = 90%
4. 0.5 = 50%
5. 0.25 = 25%

★ 0.75 = 75%

# Examples 3 and 4 — Fractions as Decimals

EXAMPLES · 5 MIN

SILVER

*Watch and copy into your book*

**Hint** Same × top and bottom  Or divide top ÷ bottom

## Example 3

Write as a decimal

$\frac{3}{4}$

## Example 4

Write as a decimal

$\frac{3}{5}$

# Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

*Check your copy and correct in green pen*

## Example 3

**Make the bottom 100**

$$\frac{3}{4} = \frac{3 \times 25}{4 \times 25}$$

$$= \frac{75}{100}$$

$$= 0.75$$

Tip: or work out 3 ÷ 4.

## Example 4

**Make the bottom 10**

$$\frac{3}{5} = \frac{3 \times 2}{5 \times 2}$$

$$= \frac{6}{10}$$

$$= 0.6$$

Tip: same × top and bottom.

# Task 2

PRACTICE · 9 MIN

SILVER

*Write as a decimal, then a percentage*

**Hint** Same × top and bottom  Or divide top ÷ bottom

1. $\frac{4}{5}$

2. $\frac{3}{20}$

3. $\frac{12}{16}$

4. $\frac{2}{5}$

5. $\frac{13}{20}$

★ $\frac{7}{8}$

# Task 2 — Answers

CORRECTIONS

SILVER

*Tick or correct in green pen*

1. 0.8 = 80%
2. 0.15 = 15%
3. 0.75 = 75%
4. 0.4 = 40%
5. 0.65 = 65%

★ 0.875 = 87.5%

# Examples 5 and 6 — Mixed Numbers and Comparing

EXAMPLES · 5 MIN

GOLD

*Exam-style: show all your working*

**Hint** 1 whole = 100%  Compare like with like

## Example 5

(2 marks)

Write as a decimal and a percentage

$1\ \frac{3}{5}$

## Example 6

(2 marks)

Which is greater?

$\frac{13}{20}$, 0.6 or 62%

# Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

*Check your copy and correct in green pen*

## Example 5

**Whole + the fraction**

$$1\ \frac{3}{5} = 1 + 0.6$$

$$= 1.6$$

$$\mathbf{= 160\%}$$

Tip: 1 whole = 100%.

## Example 6

**Change them all to percentages**

$$\frac{13}{20} = \frac{65}{100} = 65\%$$

$$0.6 = 60\%$$

$$\mathbf{\frac{13}{20}\ is\ the\ greatest.}$$

Tip: compare like with like.

# Task 3

PRACTICE · 9 MIN

GOLD

*Exam-style: convert and compare*

**Hint** 1 whole = 100%  Compare like with like

**1** $1\frac{1}{2}$

**2** $1\frac{3}{10}$

**3** $1\frac{7}{10}$

**4** Order, smallest first:

$0.3,\ \frac{1}{4},\ 28\%$

**5** Write 1.75 as a mixed number and a percentage

★ Is $\frac{9}{20}$ closer to 40% or 50%?

# Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. 1.5 = 150%
2. 1.3 = 130%
3. 1.7 = 170%
4. $\frac{1}{4}$, 28%, 0.3
5. $1\ \frac{3}{4} = 175\%$

★ 45%: the same distance

# Plenary — Exit Ticket

PLENARY · 5 MIN

*Answer in your book. Try all three.*

**BRONZE** Write $\frac{2}{5}$ as a decimal.

**SILVER** Write 0.35 as a percentage.

**GOLD** Write $1\ \frac{1}{4}$ as a percentage.

**Reflect:** Look back at the learning objectives. Which ones can you do now?

# Plenary — Answers

PLENARY

*Tick or correct in green pen*

BRONZE **0.4**

SILVER **35%**

GOLD **1.25 = 125%**

# Exam Practice — Converting

EXAM PRACTICE · 10 MIN

BRONZE

*IGCSE exam-style: show all your working*

**Hint** Percent: out of 100 Make the bottom 100

## Question 1

(2 marks)

Write as a decimal.

**(a)** $\frac{9}{10}$ (1)

.....

**(b)** $\frac{1}{4}$ (1)

.....

## Question 2

(2 marks)

Write as a percentage.

**(a)** 0.6 (1)

..... %

**(b)** $\frac{1}{2}$ (1)

..... %

# Exam Practice — Fractions to Decimals

EXAM PRACTICE · 10 MIN

SILVER

*IGCSE exam-style: show all your working*

**Hint**
Same × top and bottom
Or divide top ÷ bottom

## Question 3

(2 marks)

Write $\frac{3}{20}$ as a decimal.

Show your working.

.....

## Question 4

(3 marks)

Complete the table.

| Fraction | | | |
|---|---|---|---|
| Decimal | 0.4 | | |
| Percentage | | 75% | 35% |

.....

# Exam Practice — Comparing Values

EXAM PRACTICE · 10 MIN

GOLD

*IGCSE exam-style: show all your working*

**Hint** 1 whole = 100%  Compare like with like

## Question 5

(2 marks)

Write $1\frac{3}{4}$ as a decimal and a %.

.....

## Question 6

(3 marks)

Sam scores $\frac{17}{20}$ of the marks in a test.

Mia scores 0.82 of the marks.

Leo scores 84%.

Who scores highest? Show working.

.....

# Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 1

**(a) 0.9** B1

**(b) 0.25** B1

Tip: make the bottom 100.

## Question 2

**(a) 60%** B1

**(b) 50%** B1

Tip: 0.6 = 60 hundredths.

# Exam Practice — Mark Scheme

CORRECTIONS

SILVER

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 3

$$\frac{3}{20} = \frac{15}{100}$$ M1

$$= 0.15$$ A1

Tip: 20 × 5 = 100.

## Question 4

$\frac{2}{5}$, 0.4, 40% B1

$\frac{3}{4}$, 0.75, 75% B1

$\frac{7}{20}$, 0.35, 35% B1

Tip: one mark for each column.

# Exam Practice — Mark Scheme

CORRECTIONS

GOLD

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 5

**1.75** B1

**175%** B1

Tip: 1 whole = 100%.

## Question 6

$\frac{17}{20} = \frac{85}{100} = 85\%$ M1

Mia 82%, Leo 84% M1

**Sam** A1

Tip: change all to percentages.